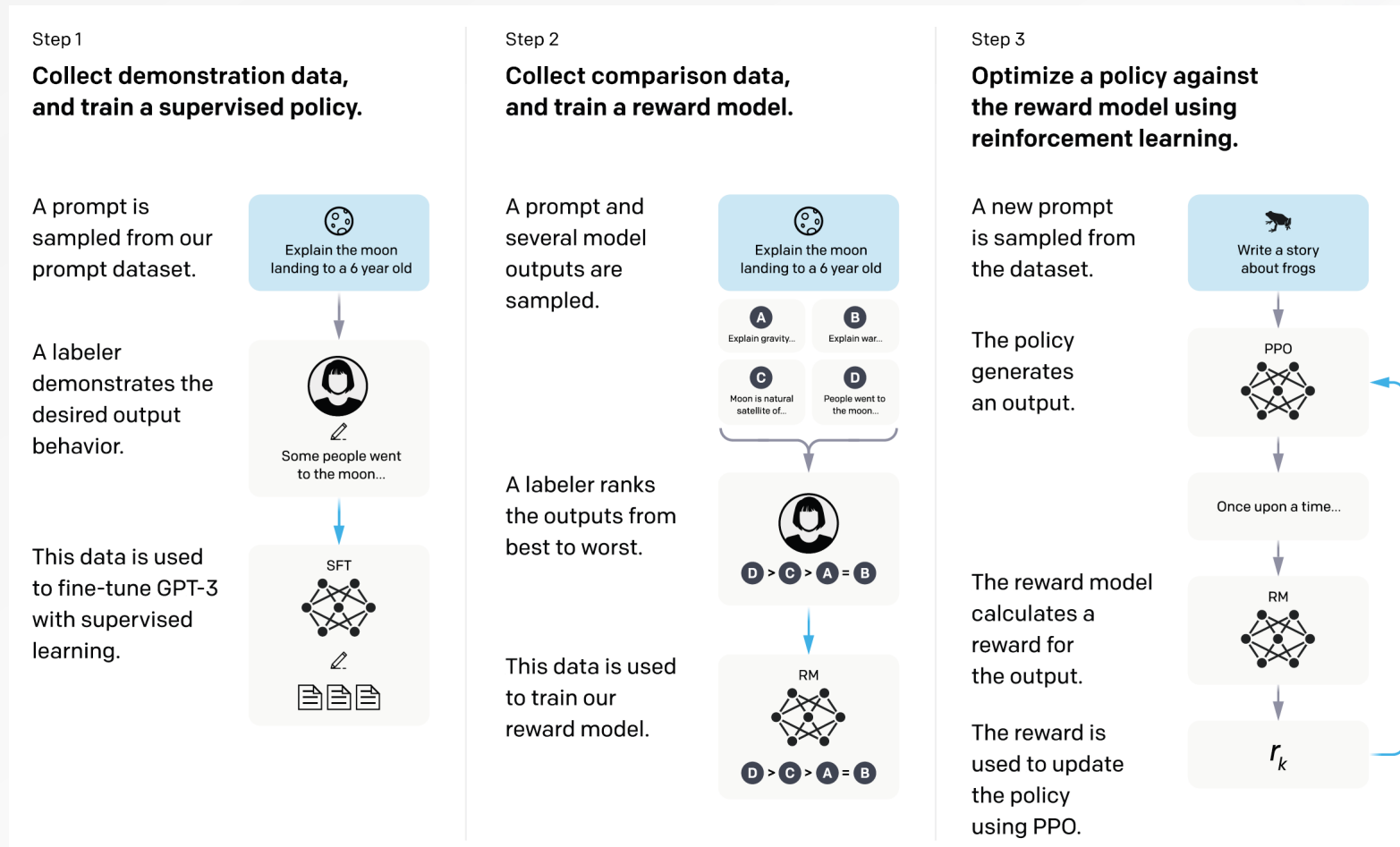


Policy Optimization in RLHF: From PPO to Recent Advances

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Reinforcement Learning from Human Feedback



(1) supervised fine-tuning (SFT), (2) reward model (RM) training, (3) reinforcement learning via proximal policy optimization (PPO) on this reward model.

Proximal Policy Optimization (PPO)



Maximize the following combined objective function in RL training:

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} [r_{\theta}(x, y) - \beta \log(\pi_{\phi}^{\text{RL}}(y | x) / \pi^{\text{SFT}}(y | x))] + \gamma E_{x \sim D_{\text{pretrain}}} [\log(\pi_{\phi}^{\text{RL}}(x))]$$

(InstructGPT refers to the PPO-ptx)

Reward

$$r_t = r_{\phi}(q, o_{\leq t}) - \beta \log \frac{\pi_{\theta}(o_t | q, o_{< t})}{\pi_{\text{ref}}(o_t | q, o_{< t})}$$

PPO

$$\mathcal{J}_{\text{PPO}}(\theta) = \mathbb{E}[q \sim P(Q), o \sim \pi_{\theta_{\text{old}}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \min \left[\frac{\pi_{\theta}(o_t | q, o_{< t})}{\pi_{\theta_{\text{old}}}(o_t | q, o_{< t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t | q, o_{< t})}{\pi_{\theta_{\text{old}}}(o_t | q, o_{< t})}, 1 - \varepsilon, 1 + \varepsilon \right) A_t \right]$$

A_t is the advantage, which is computed by applying Generalized Advantage Estimation (GAE)

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad \delta_l = R_l + \gamma V(s_{l+1}) - V(s_l), \quad 0 \leq \gamma, \lambda \leq 1.$$

Generalized Advantage Estimation (GAE)



The discounted value functions are given by:

$$V^{\pi, \gamma}(s_t) := \mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} \left[\sum_{l=0}^{\infty} \gamma^l r_{t+l} \right] \quad Q^{\pi, \gamma}(s_t, a_t) := \mathbb{E}_{s_{t+1:\infty}, a_{t+1:\infty}} \left[\sum_{l=0}^{\infty} \gamma^l r_{t+l} \right]$$

Advantage: $A^{\pi, \gamma}(s_t, a_t) := Q^{\pi, \gamma}(s_t, a_t) - V^{\pi, \gamma}(s_t)$

GAE is defined as the exponentially-weighted average of these k-step estimators:

$$\begin{aligned} \hat{A}_t^{(1)} &:= \delta_t^V &= -V(s_t) + r_t + \gamma V(s_{t+1}) \\ \hat{A}_t^{(2)} &:= \delta_t^V + \gamma \delta_{t+1}^V &= -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 V(s_{t+2}) \\ \hat{A}_t^{(3)} &:= \delta_t^V + \gamma \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V &= -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 V(s_{t+3}) \\ \hat{A}_t^{(k)} &:= \sum_{l=0}^{k-1} \gamma^l \delta_{t+l}^V &= -V(s_t) + r_t + \gamma r_{t+1} + \dots + \gamma^{k-1} r_{t+k-1} + \gamma^k V(s_{t+k}) \end{aligned}$$

$$\begin{aligned} \hat{A}_t^{\text{GAE}(\gamma, \lambda)} &:= (1 - \lambda) \left(\hat{A}_t^{(1)} + \lambda \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \dots \right) \\ &= (1 - \lambda) (\delta_t^V + \lambda (\delta_t^V + \gamma \delta_{t+1}^V) + \lambda^2 (\delta_t^V + \gamma \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V) + \dots) \\ &= (1 - \lambda) (\delta_t^V (1 + \lambda + \lambda^2 + \dots) + \gamma \delta_{t+1}^V (\lambda + \lambda^2 + \lambda^3 + \dots) \\ &\quad + \gamma^2 \delta_{t+2}^V (\lambda^2 + \lambda^3 + \lambda^4 + \dots) + \dots) \\ &= (1 - \lambda) \left(\delta_t^V \left(\frac{1}{1 - \lambda} \right) + \gamma \delta_{t+1}^V \left(\frac{\lambda}{1 - \lambda} \right) + \gamma^2 \delta_{t+2}^V \left(\frac{\lambda^2}{1 - \lambda} \right) + \dots \right) \\ &= \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V \end{aligned}$$

There are two notable special cases of this formula, obtained by setting $\lambda = 0$ and $\lambda = 1$.

$$\begin{aligned} \text{GAE}(\gamma, 0) : \quad \hat{A}_t &:= \delta_t &= r_t + \gamma V(s_{t+1}) - V(s_t) \\ \text{GAE}(\gamma, 1) : \quad \hat{A}_t &:= \sum_{l=0}^{\infty} \gamma^l \delta_{t+l}^V &= \sum_{l=0}^{\infty} \gamma^l r_{t+l} - V(s_t) \end{aligned}$$

Direct Preference Optimization (DPO)

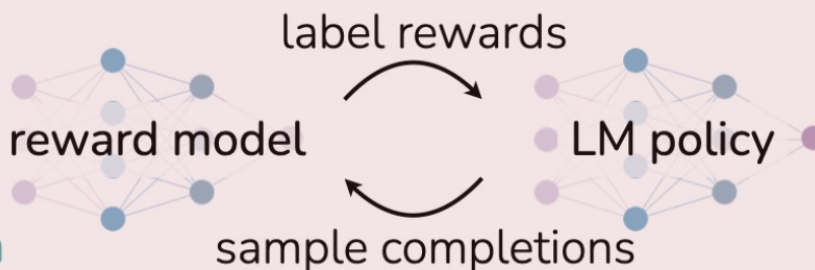
Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about
the history of jazz"



preference data

maximum
likelihood



reinforcement learning

Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



preference data

maximum
likelihood



DPO optimizes for human preferences while avoiding reinforcement learning.

- The Bradley-Terry (BT) model stipulates that the human preference distribution p^* can be written as:
- Framing the problem as a binary classification we have the negative log-likelihood loss:

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp(r^*(x, y_1))}{\exp(r^*(x, y_1)) + \exp(r^*(x, y_2))}$$

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))]$$

where σ is the logistic function $\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1}$

- RL Fine-Tuning Phase:
- The optimal solution to the KL-constrained reward maximization objective takes the form:

$$\pi_r(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right) \quad \text{where} \quad Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

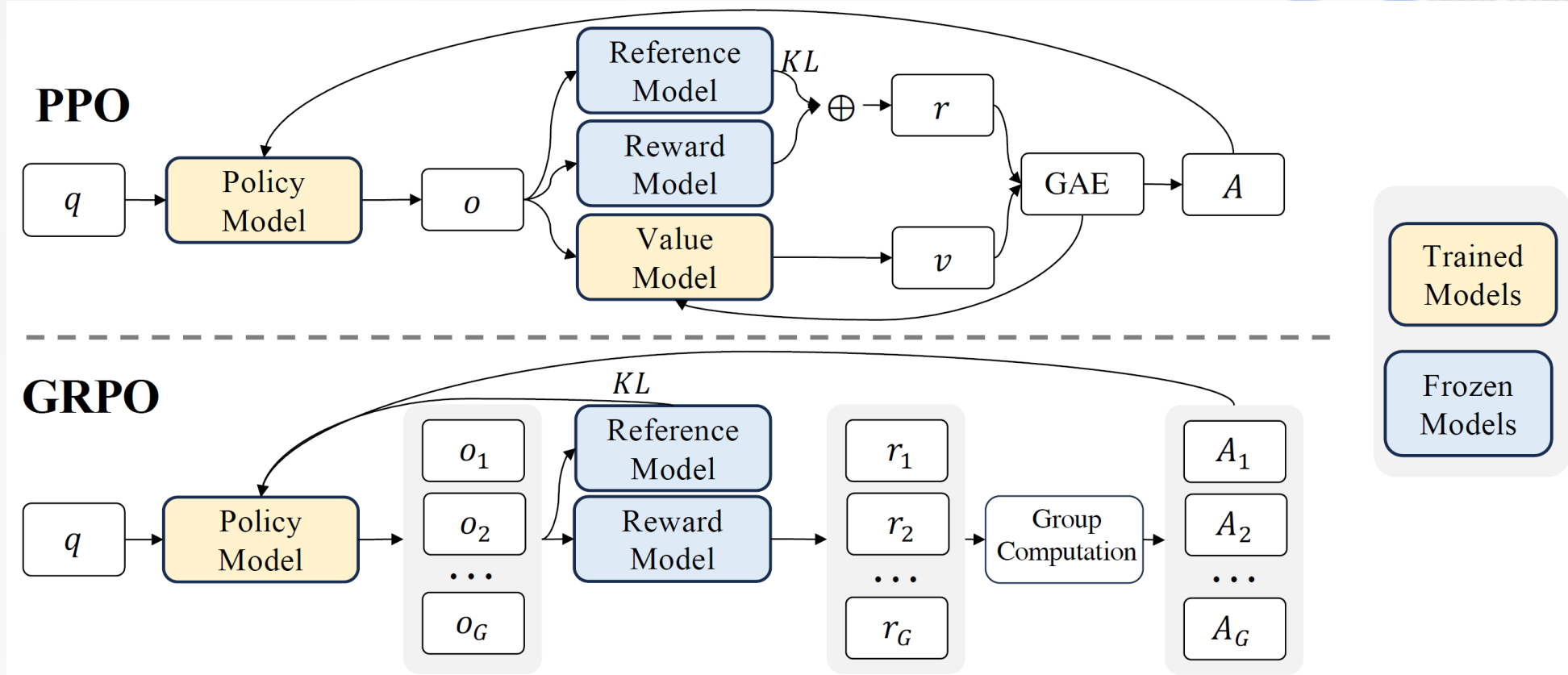
- We can express the reward function in terms of π_r , π_{ref} , and $Z(\cdot)$: $r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x)$
- Substituting the reparameterization for r into the BT model:

$$p^*(y_1 \succ y_2 \mid x) = \frac{1}{1 + \exp\left(\beta \log \frac{\pi_r(y_2 \mid x)}{\pi_{\text{ref}}(y_2 \mid x)} - \beta \log \frac{\pi_r(y_1 \mid x)}{\pi_{\text{ref}}(y_1 \mid x)}\right)}$$

- We have the probability of human preference data in terms of the optimal policy. Formulate a maximum likelihood objective for a parametrized policy π_θ :

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

Group Relative Policy Optimization (GRPO)



$$\mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \min \left[\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})}, 1 - \epsilon, 1 + \epsilon \right) A_t \right]$$

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} [\pi_{\theta} || \pi_{ref}] \right\}$$

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL} [\pi_{\theta} || \pi_{ref}] \right\}$$

- $\hat{A}_{i,t}$ is the advantage calculated based on relative rewards of the outputs inside each group only

$$\mathbf{r} = \{r_1, r_2, \dots, r_G\} \quad \hat{A}_{i,t} = \tilde{r}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$$

- GRPO regularizes by directly adding the KL divergence instead of adding KL penalty in the reward
- Estimate the KL divergence with the following unbiased estimator, which is guaranteed to be positive.

$$\mathbb{D}_{KL} [\pi_{\theta} || \pi_{ref}] = \frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - \log \frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - 1$$

REINFORCE Leave-One-Out (RLOO)



Given that in LLM applications, $r(x, y)$ is only obtained at the end of the full sequence, the modeling of partial sequences is unnecessary

REINFORCE

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [R(y, x) \nabla_{\theta} \log \pi_{\theta}(y|x)]$$

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [(R(y, x) - b) \nabla_{\theta} \log \pi_{\theta}(y|x)]$$

$$b_{\text{MA}} = \frac{1}{S} \sum_s R(x^s, y^s)$$

moving average of all rewards throughout training

RLOO: The rewards for each sample can serve all other samples as a baseline

$$\frac{1}{k} \sum_{i=1}^k [R(y_{(i)}, x) - \frac{1}{k-1} \sum_{j \neq i} R(y_{(j)}, x)] \nabla \log \pi(y_{(i)}|x) \text{ for } y_{(1)}, \dots, y_{(k)} \stackrel{i.i.d}{\sim} \pi_{\theta}(\cdot|x)$$

Decoupled Clip and Dynamic sAmpling Policy Optimization (DAPO)



- Removing KL Divergence
- Rule-based Reward Modeling

$$R(\hat{y}, y) = \begin{cases} 1, & \text{is_equivalent}(\hat{y}, y) \\ -1, & \text{otherwise} \end{cases}$$

$$\begin{aligned} \mathcal{J}_{\text{DAPO}}(\theta) = & \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \\ & \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right] \\ \text{s.t. } & 0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G, \end{aligned}$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}.$$

1. Clip-Higher, which promotes the diversity of the system and avoids entropy collapse;
2. Dynamic Sampling, which improves training efficiency and stability;
3. Token-Level Policy Gradient Loss, which is critical in long-CoT RL scenarios;
4. Overlong Reward Shaping, which reduces reward noise and stabilizes training.

- **Clip-Higher:** The upper clipping threshold restricts the probability increase of low-probability ‘exploration’ tokens. Decouple the lower and higher clipping range as ϵ_{low} and ϵ_{high} , increase the value of ϵ_{high} to leave more room for the increase of low-probability tokens

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right]$$

s.t. $0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$

- **Dynamic Sampling:** keep sampling until the batch is fully filled with samples whose accuracy is neither 0 nor 1

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right]$$

s.t. $0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$

- **Token-Level Policy Gradient Loss**

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right]$$

s.t. $0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$

- **Overlong Reward Shaping:** set a maximum length for generation, masks the loss of truncated samples; Soft Overlong Punishment

$$R_{\text{length}}(y) = \begin{cases} 0, & |y| \leq L_{\text{max}} - L_{\text{cache}} \\ \frac{(L_{\text{max}} - L_{\text{cache}}) - |y|}{L_{\text{cache}}}, & L_{\text{max}} - L_{\text{cache}} < |y| \leq L_{\text{max}} \\ -1, & L_{\text{max}} < |y| \end{cases}$$

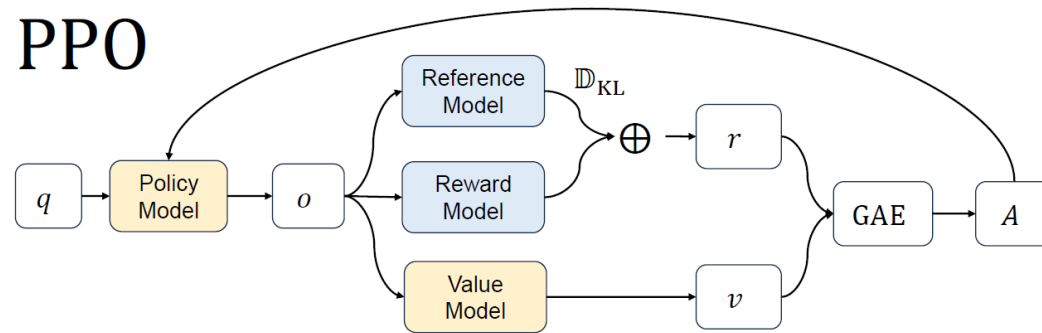
$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right]$$

where

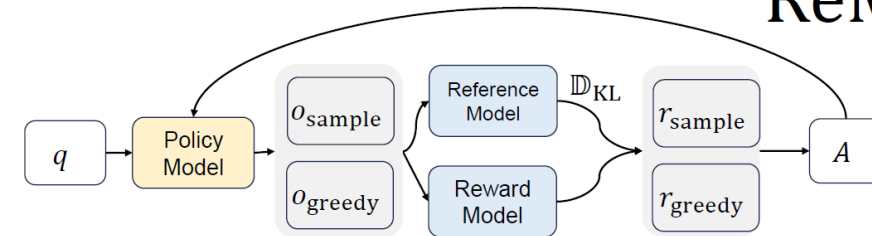
$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}.$$

REINFORCE++

PPO

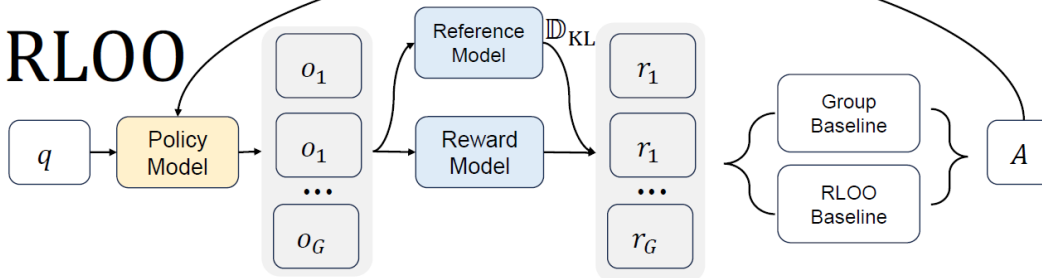


ReMax



GRPO

RLOO



Reinforce++

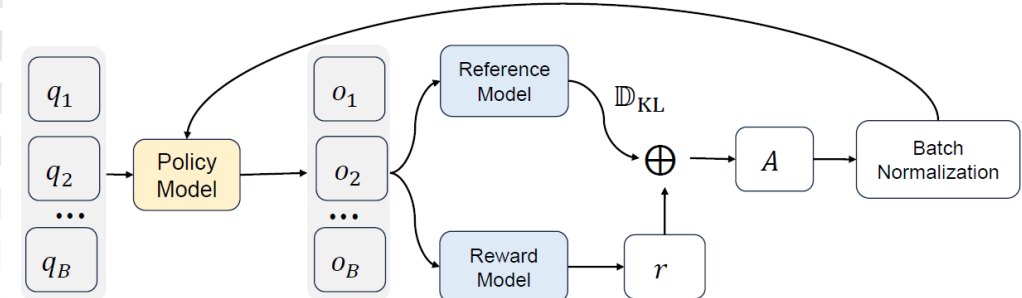


Figure 1: The comparison of PPO, Remax, GRPO, RLOO, and REINFORCE++.

They estimate advantages independently for responses to each prompt, which can lead to overfitting on simpler prompts and vulnerability to reward hacking and may be biased

REINFORCE++

- Adopt the average reward of a global training batch as the baseline reward

$$A_{q,o_t} = r(o_{1:T}, q) - \beta \cdot \sum_{i=t}^T \text{KL}(i)$$

- Normalize this advantage across the global batch for all prompts

$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}(A_{q,o_t} \mid A_{q,o_t} \in \mathcal{D}_{\text{batch}})}{\text{std}(A_{q,o_t} \mid A_{q,o_t} \in \mathcal{D}_{\text{batch}})}$$

- REINFORCE++-Baseline: which integrates REINFORCE++ with multiple-response generation.

$$A_{q,o_t} = R_{q,o_t} - \text{mean}_{\text{group}}(R_{q,o_t})$$
$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}_{\text{batch}}(A_{q,o_t})}{\text{std}_{\text{batch}}(A_{q,o_t})}$$

- REINFORCE++ still optimizes the PPO objective and employs the clipping strategy

GRPO Done Right (Dr. GRPO)

GRPO

$$\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] \right\},$$

where $\hat{A}_{i,t} = \frac{R(\mathbf{q}, \mathbf{o}_i) - \text{mean}(\{R(\mathbf{q}, \mathbf{o}_1), \dots, R(\mathbf{q}, \mathbf{o}_G)\})}{\text{std}(\{R(\mathbf{q}, \mathbf{o}_1), \dots, R(\mathbf{q}, \mathbf{o}_G)\})}.$

Dr. GRPO

GRPO Done Right (without bias)

$$\frac{1}{G} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|\mathbf{q}, \mathbf{o}_{i,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right] \right\},$$

where $\hat{A}_{i,t} = R(\mathbf{q}, \mathbf{o}_i) - \text{mean}(\{R(\mathbf{q}, \mathbf{o}_1), \dots, R(\mathbf{q}, \mathbf{o}_G)\}).$

simply remove $1/|o_i|$ and $\text{std}(\{R(q, o_1), \dots, R(q, o_G)\})$

- Response-level length bias caused by dividing $|o_i|$.
 - $\hat{A}_{i,t} > 0$, greater gradient updates for shorter responses, leading the policy to favor brevity in correct answers.
 - $\hat{A}_{i,t} < 0$, longer responses are penalized less, causing the policy to prefer lengthier responses among incorrect ones.
- Question-level difficulty bias caused by dividing $\text{std}(\{R(q, o_1), \dots, R(q, o_G)\})$.
Questions with lower standard deviations (too easy or too hard) are given higher weights during policy updates

Group Sequence Policy Optimization (GSPO)



Since the reward is granted to the entire sequence, applying off-policy correction at the token level appears problematic

- GSPO defines the importance ratio based on sequence likelihood and performs sequence-level clipping, rewarding, and optimization

$$\mathcal{J}_{\text{GSPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip} \left(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_i \right) \right]$$

$$s_i(\theta) = \left(\frac{\pi_{\theta}(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_{\theta}(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right)$$

- Group-based advantage estimation:

$$\hat{A}_i = \frac{r(x, y_i) - \text{mean}(\{r(x, y_i)\}_{i=1}^G)}{\text{std}(\{r(x, y_i)\}_{i=1}^G)}$$

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)}$$

$$\left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} || \pi_{\text{ref}}) \right) \right]$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} | q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} | q, o_{i,<t})}.$$

Summary

Algorithm	#Models	KL	Query-Answer	Novelty
PPO	4(policy,value,ref,reward)	in reward	1-1	First RLHF
DPO	2(policy,ref)	in reward	1-2	No RL
ReMax	3(policy,ref,reward)	in reward	1-2	Greedy advantage
GRPO	3(policy,ref,reward)	single term	1-G	Group relative advantage
RLOO	3(policy,ref,reward)	in reward	1-G	REINFORCE, Sequence-level optimization
DAPO	3(policy,ref,reward)	No KL	1-G	Clip, Sampling, Token-Level PG, Reward Shaping
REINFORCE++	3(policy,ref,reward)	in reward	B-B B-BG	Global (prompt) advantage normalization
Dr. GRPO	3(policy,ref,reward)	single term	1-G	Remove $1/ o_i $ and $\text{std}(\{R(q, o_1), \dots, R(q, o_G)\})$
GSPO	3(policy,ref,reward)	single term	1-G	Sequence-level importance sampling

Next?

Tokenizer: BPE, Word-Piece

Embedding: fixed, learning

Position embedding: Sinusoidal, Learned, RoPE

Optimizer: SGD, SGDM, Adagrad, RMSProp, Adam, Adamw, Muon, MuonClip

Finetune: LoRA, QLoRA, Adapter Tuning, Prefix Tuning, Prompt Tuning, P-Tuning

Activation function:

ReLU

$$A(x) = \max(x, 0)$$

GELU

$$\text{GELU}(x) = x \cdot \Phi(x) = x \cdot \frac{1}{2} \left[1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right]$$

Swish (SiLU)

$$\text{swish}(x) = x \cdot \sigma(x)$$

GLU

$$\text{GLU}(x) = (XW + b) \otimes \sigma(XV + c)$$

SwiGLU

$$\text{SwiGLU}(x) = (XW + b) \otimes \text{Swish}(XV + c)$$

GeGLU

$$\text{GeGLU}(x) = (XW + b) \otimes \text{GELU}(XV + c)$$

Reranker, MoE ...



THANKS!

